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Lab Preview

SUREN NETWORK
PLAY WITH TECHNOLOGIES

```
graph LR; Win1[Win1 50.0.0.10] --- e0((e0)) --- Router[Router 50.0.0.1]; Router --- e0_1((e0/1)) --- Win2[Win2 60.0.0.10];
```

5. LAB Routing
By Meganadhan M A

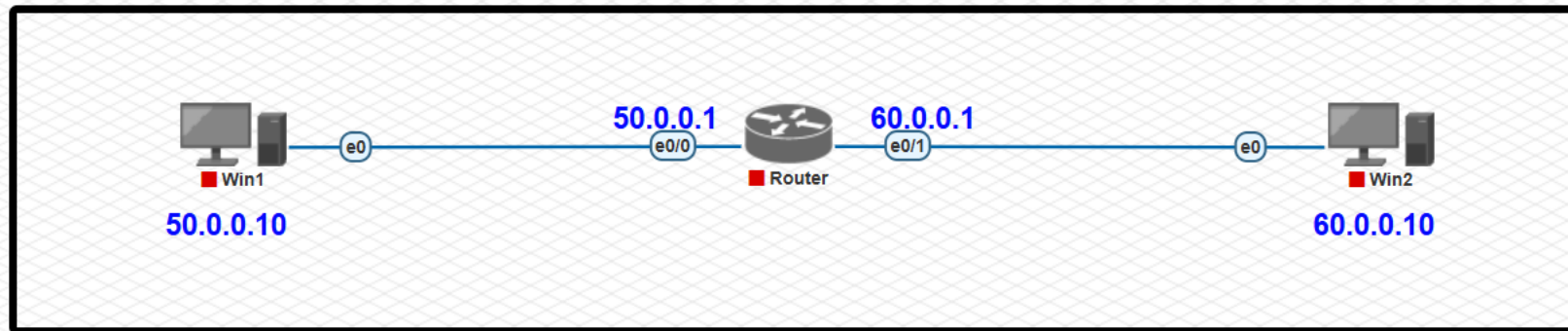
Open Edit Delete

Lab Name : 5-LAB ROUTING

Author : 1

Version : 1

LAB ROUTING TOPOLOGY



labs.surennetwork.com/store/public/admin/labs/terminal?terminals=eyJ0YWJzljpbeyJuYW1lIjoiv2luMSIsInVy...

Win1 x Router x Win2 x

```
Router#sh ip int br
Interface          IP-Address      OK? Method Status      Protocol
Ethernet0/0        unassigned      YES TFTP   up          up
Ethernet0/1        unassigned      YES TFTP   up          up
Ethernet0/2        unassigned      YES TFTP   down        down
Ethernet0/3        unassigned      YES TFTP   down        down

Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int e 0/0
Router(config-if)#ip add 50.0.0.1 255.0.0.0
Router(config-if)#no shut
Router(config-if)#int e 0/1
Router(config-if)#ip add 60.0.0.1 255.0.0.0
Router(config-if)#no shut
Router(config-if)#ex
Router(config)#username Admin password mkv@10
Router(config)#line con 0
Router(config-line)#password mkv@10
Router(config-line)#login local
Router(config-line)#exit
Router(config)#
*Sep  3 11:46:03.349: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
Router(config)#enable secret mkv
Router(config)#
Router(config)#line vty 0 5
^
% Invalid input detected at '^' marker.

Router(config)#line vty 0 4
Router(config-line)#transport input telnet
Router(config-line)#password mkv@30
Router(config-line)#login local
Router(config-line)#exit
Router(config)#
```

Router Configuration direct interface connection

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05:18 pm 03-09-2026

Win1 × Router × Win2 ×

Ethernet0/1	60.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```
Router(config)#do sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```

50.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    50.0.0.0/8 is directly connected, Ethernet0/0
L    50.0.0.1/32 is directly connected, Ethernet0/0
60.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    60.0.0.0/8 is directly connected, Ethernet0/1
L    60.0.0.1/32 is directly connected, Ethernet0/1

```

```
Router(config)#do sh ip route static
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```
Router(config)#
```

Router Configuration direct interface connection

labs.surennetwork.com/store/public/admin/labs/terminal?terminals=eyJ0YWJzljpbeyJuYW1lIjoivV2luMSIsInVy...

Win1 x Router x Win2 x

Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms
Router(config)#do ping 60.0.0.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 60.0.0.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms
Router(config)#do ping 60.0.0.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 60.0.0.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/4/5 ms
Router(config)#do sh ip int br

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	50.0.0.1	YES	manual	up	up
Ethernet0/1	60.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

Router(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

50.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 50.0.0.0/8 is directly connected, Ethernet0/0
L 50.0.0.1/32 is directly connected, Ethernet0/0
60.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 60.0.0.0/8 is directly connected, Ethernet0/1
L 60.0.0.1/32 is directly connected, Ethernet0/1

Router Configuration direct interface connection

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05:20 pm 03-09-2026

labs.surennetwork.com/store/public/admin/labs/terminal?terminals=eyJ0YWJzljpbeyJuYW1lloiV2luMSIsInVy...

linkedinWebmail - MainNetworking Acade...Online typing WPM...NCCR Biometric Att...HP Product Warrant...Fujitsu Technical Su...D-LINKHP » Device StatusAll Bookmarks

Win1 xRouter xWin2 x

C:\WINDOWS\system32\cmd.exe

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . :
IP Address : 50.0.0.10
Subnet Mask : 255.0.0.0
Default Gateway :

C:\>ping 50.0.0.10

Pinging 50.0.0.10 with 32 bytes of data:

Reply from 50.0.0.10: bytes=32 time<1ms TTL=128
Reply from 50.0.0.10: bytes=32 time<1ms TTL=128
Reply from 50.0.0.10: bytes=32 time<1ms TTL=128
Reply from 50.0.0.10: bytes=32 time<1ms TTL=128

Ping statistics for 50.0.0.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 50.0.0.1

Pinging 50.0.0.1 with 32 bytes of data:

Reply from 50.0.0.1: bytes=32 time=1ms TTL=255
Reply from 50.0.0.1: bytes=32 time=1ms TTL=255
Reply from 50.0.0.1: bytes=32 time=1ms TTL=255
Reply from 50.0.0.1: bytes=32 time=1ms TTL=255

Ping statistics for 50.0.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 60.0.0.10

Pinging 60.0.0.10 with 32 bytes of data:

Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.

Ping statistics for 60.0.0.10:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 60.0.0.1

Pinging 60.0.0.1 with 32 bytes of data:

Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.

Ping statistics for 60.0.0.1:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

Before adding the Gateway to the PC

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03-09-2026


```
C:\WINDOWS\system32\cmd.exe
Subnet Mask . . . . . : 255.0.0.0
Default Gateway . . . . . : 60.0.0.1

C:\>ping 60.0.0.1

Pinging 60.0.0.1 with 32 bytes of data:

Reply from 60.0.0.1: bytes=32 time=1ms TTL=255
Reply from 60.0.0.1: bytes=32 time=1ms TTL=255
Reply from 60.0.0.1: bytes=32 time=1ms TTL=255
Reply from 60.0.0.1: bytes=32 time=1ms TTL=255

Ping statistics for 60.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 60.0.0.10

Pinging 60.0.0.10 with 32 bytes of data:

Reply from 60.0.0.10: bytes=32 time<1ms TTL=128
Reply from 60.0.0.10: bytes=32 time<1ms TTL=128
Reply from 60.0.0.10: bytes=32 time<1ms TTL=128
Reply from 60.0.0.10: bytes=32 time<1ms TTL=128

Ping statistics for 60.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 50.0.0.10

Pinging 50.0.0.10 with 32 bytes of data:

Reply from 50.0.0.10: bytes=32 time=2ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127

Ping statistics for 50.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 50.0.0.1

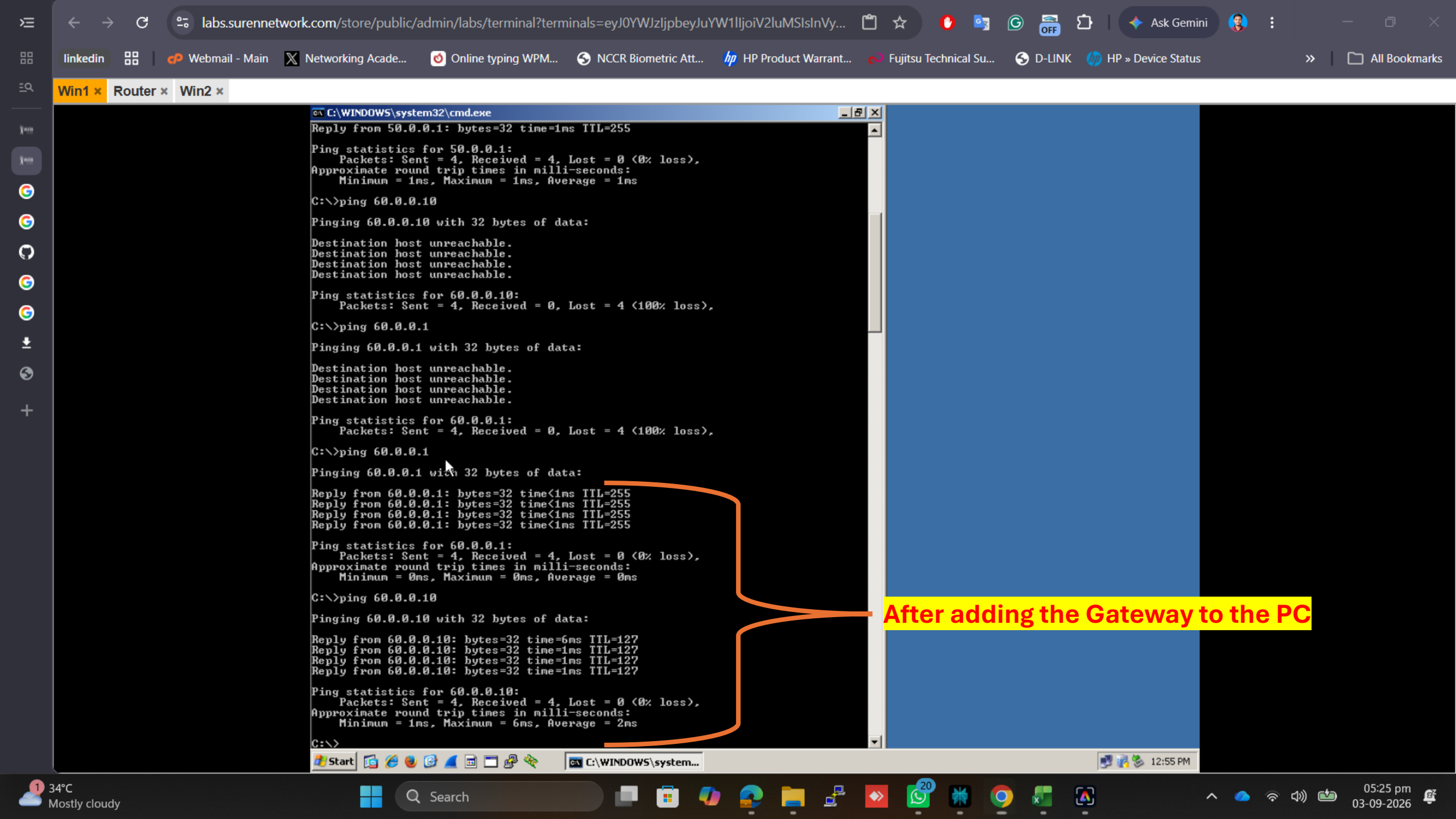
Pinging 50.0.0.1 with 32 bytes of data:

Reply from 50.0.0.1: bytes=32 time<1ms TTL=255
Reply from 50.0.0.1: bytes=32 time=1ms TTL=255
Reply from 50.0.0.1: bytes=32 time=1ms TTL=255
Reply from 50.0.0.1: bytes=32 time=1ms TTL=255

Ping statistics for 50.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

After adding the Gateway to the PC



```
C:\WINDOWS\system32\cmd.exe
Reply from 50.0.0.1: bytes=32 time=1ms TTL=255
Ping statistics for 50.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\>ping 60.0.0.10
Pinging 60.0.0.10 with 32 bytes of data:
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Ping statistics for 60.0.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 60.0.0.1
Pinging 60.0.0.1 with 32 bytes of data:
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Ping statistics for 60.0.0.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 60.0.0.1
Pinging 60.0.0.1 with 32 bytes of data:
Reply from 60.0.0.1: bytes=32 time<1ms TTL=255
Reply from 60.0.0.1: bytes=32 time<1ms TTL=255
Reply from 60.0.0.1: bytes=32 time<1ms TTL=255
Reply from 60.0.0.1: bytes=32 time<1ms TTL=255
Ping statistics for 60.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 60.0.0.10
Pinging 60.0.0.10 with 32 bytes of data:
Reply from 60.0.0.10: bytes=32 time=6ms TTL=127
Reply from 60.0.0.10: bytes=32 time=1ms TTL=127
Reply from 60.0.0.10: bytes=32 time=1ms TTL=127
Reply from 60.0.0.10: bytes=32 time=1ms TTL=127
Ping statistics for 60.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 6ms, Average = 2ms
C:\>
```

After adding the Gateway to the PC